



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CAPSTONE PROJECT PRESENTATION

PROJECT TITLE

Automatic Text Summarization Using a Transformer-Based T5 Model

PRESENTED BY

Pradeepkumar (192425016)

Yuvaraj R (192524058)

G. Jaswanth (192472235)

GUIDED BY

Dr. Kumaragurubaran T



Introduction



Overview

Automatic text summarization converts lengthy documents into concise, meaningful summaries. The proposed system uses the transformer-based T5 (Text-to-Text Transfer Transformer) model to generate abstractive summaries by understanding context and producing coherent, natural-language text.

Problem Statement

Traditional summarization methods mainly extract existing sentences and may fail to preserve the meaning, coherence, and context of complex documents. This makes them less effective for large and information-rich text.

Purpose

To develop an automatic text summarization system using the T5 model that generates accurate, concise, fluent, and context-aware summaries for large documents.



Objectives



Define Clear Goals

To design and implement an automatic text summarization system using a transformer-based T5 model.

Address the Problem Statement

To overcome the limitations of traditional summarization methods by generating coherent, context-aware, and meaningful abstractive summaries.

Expected Outcomes

To develop an efficient summarization model that produces accurate, concise, and grammatically correct summaries for large text documents.



Background



Existing Research

Early text summarization approaches primarily focused on extractive techniques, while recent research increasingly uses transformer-based models for abstractive summarization. Models such as BERT, GPT, and T5 have demonstrated improved performance by capturing contextual and semantic relationships through attention mechanisms.

Research Gap

Despite these advancements, challenges remain in generating concise and coherent summaries for long documents while minimizing information loss. This highlights the need for optimized T5-based approaches that can improve summary quality and handle long-document summarization effectively.



Methodology



Tools & Frameworks

The system is developed using Python with deep learning and NLP libraries such as PyTorch, Hugging Face Transformers, and tokenizer utilities. Pandas and NumPy are used for data handling and analysis, while ROUGE metrics are used for model evaluation.

Development Approach

A transformer-based abstractive summarization approach is adopted. A pretrained T5/mT5 model is fine-tuned on multilingual data using a supervised learning pipeline consisting of preprocessing, training, validation, and testing.

Data Collection

The dataset consists of multilingual news articles and their human-written summaries collected from online news sources. The data is organized into training, validation, and test sets to support effective model training and evaluation.

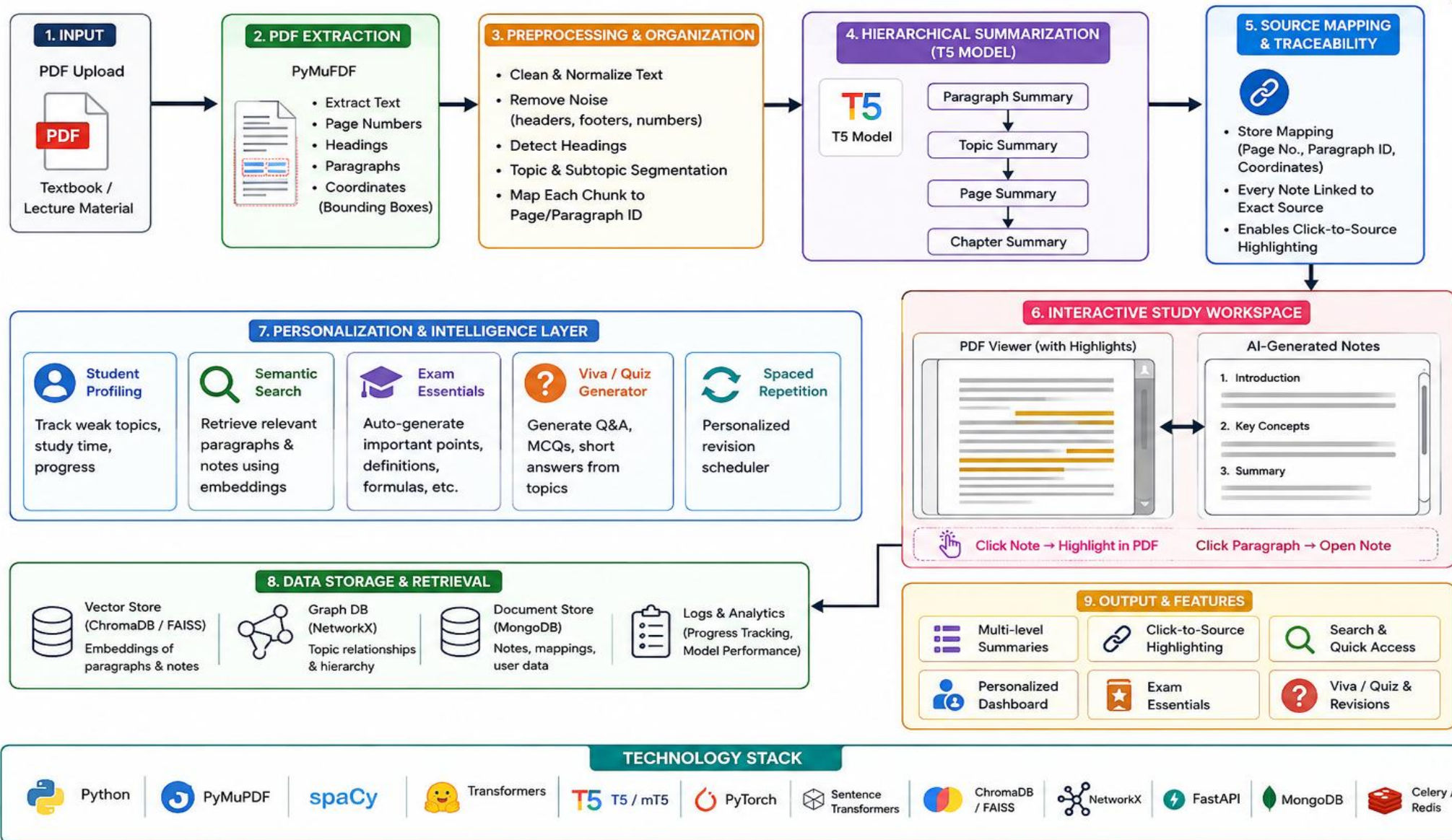


Engineer to Excel



SYSTEM ARCHITECTURE

T5-Based Automatic Text Summarization & Traceable Study Platform



Module 1: Data Collection, Preparation and Text Preprocessing

Description

- Collection of text datasets from reliable sources.
- Cleaning text by removing noise, special characters, and unnecessary symbols.
- Tokenization, normalization, and dataset formatting for T5 model training and evaluation.
- Splitting the dataset into training, validation, and testing sets.

Advantages

- Improves data quality and consistency.
- Reduces noise in raw text.
- Enhances model training efficiency and accuracy.
- Produces a well-structured dataset for summarization.

Limitations

- Over-cleaning may remove useful contextual information.
- Preprocessing large datasets can be time-consuming.
- Preprocessing rules may vary across different languages and domains.

Raw Text Data

Text Cleaning

Tokenization

Normalization

Preprocessed
Dataset

Module 2: Transformer-Based T5 Model Development & Summarization



Description

- Load the pre-trained T5 Transformer model.
- Fine-tune the model using the prepared dataset.
- Configure hyperparameters such as learning rate, batch size, and epochs.
- Generate abstractive summaries using the fine-tuned T5 model.
- Save the trained model for summary generation.

Advantages

- Produces high-quality abstractive summaries.
- Utilizes transfer learning, reducing training requirements.
- Provides strong contextual understanding for summarization.
- Generates concise and human-like summaries.

Limitations

- Requires high computational resources such as GPU/TPU.
- Fine-tuning can be time-consuming.
- Performance depends heavily on dataset quality.

Preprocessed Text

T5 Encoder

Attention
Mechanism

T5 Decoder

Generated
Summary



Engineer to Excel



Module 3: Performance Evaluation, Web Integration & Visualization

Description

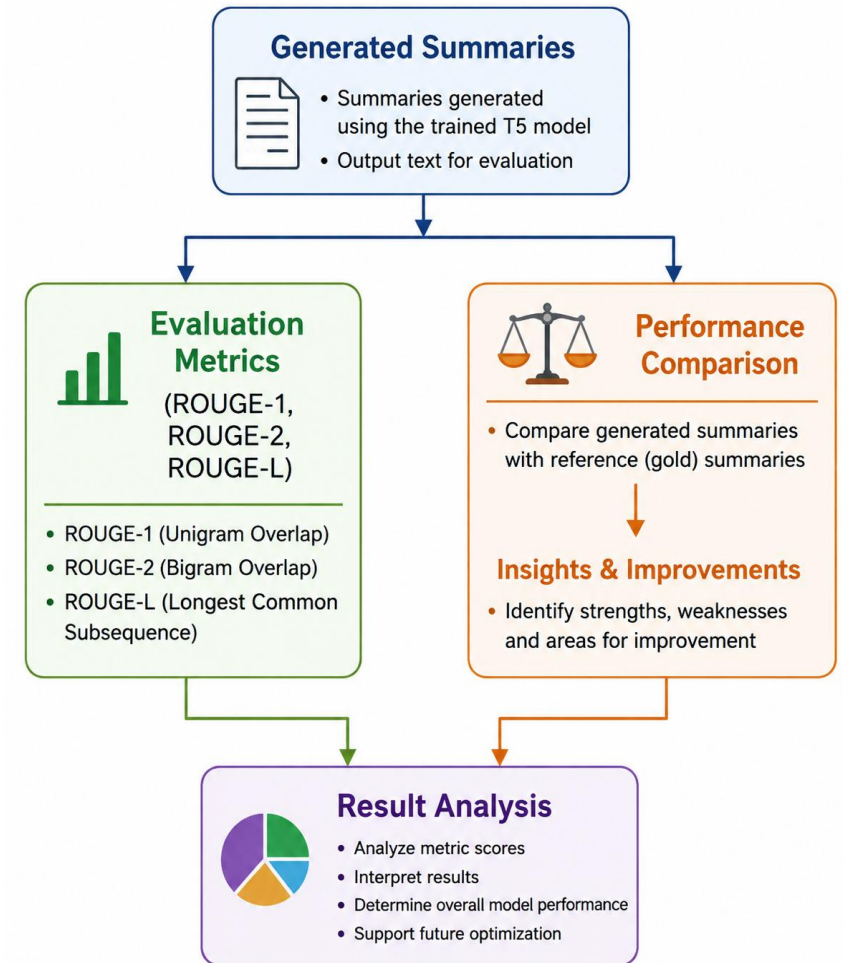
- Generate summaries using the trained T5 model.
- Compare generated summaries with reference summaries.
- Evaluate model performance using ROUGE-1, ROUGE-2, and ROUGE-L metrics.
- Analyze summary quality and identify areas for performance improvement.

Advantages

- Provides objective evaluation of summarization quality.
- Measures performance using standard NLP evaluation metrics.
- Helps identify the strengths and weaknesses of the model.
- Supports further performance optimization.

Limitations

- ROUGE focuses mainly on word overlap rather than complete semantic meaning.
- Different evaluation metrics may produce varying results.
- Human evaluation is still required for a complete assessment of summary quality.





Description About The Datasets Used

Link to Dataset

→ [MLSUM Multilingual Summarization Dataset – Kaggle](#)

The MLSUM (Multilingual Summarization Dataset) is a large multilingual news summarization dataset created from online newspapers. It contains more than 1.5 million article–summary pairs across multiple languages, making it suitable for training and evaluating automated text summarization models.

Key Points

- Includes news articles with human-written summaries.
- Covers five major languages: French, German, Spanish, Russian, and Turkish.
- Designed for multilingual NLP research, particularly abstractive summarization.
- Organized into training, validation, and testing splits for model development and evaluation.

Applications of the Research

- **Automatic News Summarization:** Quickly generate concise summaries of multilingual news articles for faster information consumption.
- **Information Retrieval Systems:** Improve search engines and digital libraries by providing brief, relevant content summaries.
- **Media Monitoring & Analysis:** Assist journalists and analysts in tracking trends and key events across multiple languages.
- **Educational Platforms:** Help students and researchers understand lengthy documents efficiently.
- **Business Intelligence:** Enable organizations to analyze large volumes of multilingual textual data for decision-making.
- **Cross-Lingual Communication:** Support summarization in different languages, enhancing global accessibility of information.

Comparison & Analysis

Aspect	Traditional Methods	Proposed T5 Model
Approach	Extractive	Abstractive
Language Support	Single language	Multilingual
Context Understanding	Limited	Strong (Transformer-based)
Summary Quality	Less fluent	More coherent & natural
Scalability	Low	High

Conclusion

- Implemented a **Transformer-based T5 model** for multilingual text summarization
- Utilized the **MLSAM multilingual news dataset**
- Generated **concise and coherent abstractive summaries**
- Achieved better performance than **traditional summarization methods**
- Demonstrated effectiveness of **pretrained multilingual models**
- Suitable for real-world **multilingual NLP applications**

Future Enhancements

- Extend the system to support **additional low-resource and regional languages**, improving inclusivity and global accessibility.
- Adapt the summarization model to work across **multiple domains** such as medical, legal, academic, and social media texts.
- Enhance performance on **long documents** by integrating long-context Transformer architectures or hierarchical summarization techniques.
- Incorporate **human-in-the-loop evaluation** to refine summary quality based on user feedback and expert judgment.
- Improve computational efficiency through **model optimization, pruning, or distillation** for faster and cost-effective deployment.
- Deploy the system as a **scalable web or mobile application** with real-time summarization capabilities for practical use.

References

- J. Wang, F. Meng, D. Zheng, Y. Liang, Z. Li, J. Qu, and J. Zhou, “Towards Unifying Multi-Lingual and Cross-Lingual Summarization with Many-to-Many Models,” arXiv Preprint, May 2023.
- E. Daraghmi, L. Atwe, and A. Jaber, “A Comparative Study of PEGASUS, BART, and T5 for Text Summarization Across Diverse Datasets,” Future Internet, vol. 17, no. 9, 2025.
- R. Krishna, K. M. Somasundaram, P. Arulmozhivarman, et al., “Deep learning for text summarization using NLP for automated news digest,” Sci. Rep., vol. 15, 36343, 2025.
- P. Dhakal and D. S. Baral, “Abstractive Summarization of Low-Resourced Nepali Language Using Multilingual Transformers,” Proc. CHIPSAL 2025, pp. 124–133, 2025.
- M. Azhar, A. Amjad, G. Farid, D. A. Dewi, and M. Batumalay, “Efficient Transformer-Based Abstractive Urdu Text Summarization Through Selective Attention Pruning,” Information, vol. 16, no. 11, 991, 2025.



Engineer to Excel



THANK YOU